



The University of Lahore

Department of Computer Science & IT

HealSearch

Fall 2021 – Spring 2025

Abstract

HealSearch is a cross-platform application designed to streamline pharmacy and medical store operations.

Objectives:

- Digitize inventory, sales, and reporting for pharmacies.
- Provide real-time data access and management.
- Enhance user support with an AI-powered assistant.

Functionalities:

- Inventory and product management
- Order processing and invoice generation
- User registration and profile management
- Sales and inventory reporting
- AI chatbot for instant help
- Location-based features (mobile)

Motivation:

Manual pharmacy management is error-prone and inefficient. HealSearch offers a modern, accessible solution to improve accuracy, efficiency, and customer service.

System Architecture



Technologies Used:

- Frontend: Flutter (Dart)
- Backend/Database: Firebase (Firestore, Auth, Storage)
- AI/ML: Gemini API
- Other: Google Maps API, Provider (state management), QuickAlert, Shared Preferences

Deployment:

- Desktop: Windows
- Mobile: Android/iOS
- Cloud: Firebase

Tools & Technologies

- **Language:** Dart
- **Framework:** Flutter
- **Database:** Firebase Firestore
- **Authentication & Storage:** Firebase Auth, Firebase Storage
- **State Management:** Provider
- **AI/ML:** Gemini API
- **Other Libraries:** QuickAlert, Google Fonts, Shared Preferences, Geolocator
- **Version Control:** Git (GitHub)

Supervisor Name: Ma'am Aqsa Aqdas

Group Members:

Members 1 : Talha Aslam

Members 2 : Muhammad Hamza Farid